

Five key factors that lead to electronic systems failing

- Electromagnetic interference
- Vibration
- Heat
- PCB assembly faults
- PCB fabrication

portant factor. Hence, when compared to the PCBs made for the consumer sector, these are made with much better quality material. Also, they have tighter electrical specifications and are fabricated using controlled fabrication techniques.

High-reliability boards are mainly used in the military and aeronautical sectors for strategic applications. These PCBs cannot afford to fail under extreme conditions and critical applications. They must provide the best performance, have extremely good electrical properties and a high-quality base material. They must be produced in a tightly controlled manufacturing process and should be able to work in harsh environments.

The different environments that spacecraft PCBs must operate in

PCBs in spacecraft basically experience three kinds of environments—ground, launch and the space environment. In the ground environment, i.e., during testing and fabrication, the PCB undergoes a lot of cleaning, thermal cycling, testing, soldering and other such processes. During the launch stage, there is heavy vibration due to the yawing and pitch spinning of the launch vehicle. Consequently, there is a lot of acoustic noise that is transferred to the structure of the vehicle, which is passed onto the electronic components. PCBs supporting such electronic packages should be able to quench the vibrations such that they are not transmitted to the components, causing damage to the leads.

These are important factors to be kept in mind while designing PCBs for launch vehicle programmes or any

space programme. Once the launch is successful, the spacecraft has to withstand the forces of the space environment, which are much harsher than the ground or launch environment. The electronic packages incorporated on to PCBs used in spacecraft will face a lot of radiation and sudden changes in temperature.

Characteristics to be measured to ensure quality

Since it is virtually impossible to rework on a PCB that has been manufactured, the process qualification plays a major role in ensuring a reliable PCB. Process qualification is basically an assurance of quality in manufacturing processes and testing methods. It confirms that products with predetermined quality characteristics and attributes can be reproduced consistently within the established limits of the manufacturing process.

Mechanical aspects. It is important to measure the length, breadth and thickness of PCBs while designing them. Pitches are verified for mounting because once the components are mounted, the mechanical characteristics of the PCB cannot be altered. Warp and twist play an important role during assembly. Flexural strength, bond strength, peel strength, trace width, thickness, resolution, protective coatings (solder mask, solder coating, gold plating, etc), dielectric thickness, etch back, barrel regularity and annular rings are the other aspects that are verified before subjecting PCBs to environmental tests.

Electrical aspects. Intra-layer and inter-layer insulation checks are done because up to 14 layers are qualified for space applications. Hence, the intra-layer and inter-layer insulation resistances, dielectric withstanding capabilities and interconnection resistances play an important role. The current carrying capacity is another important characteristic to be measured.

After mechanical and electrical checks, the PCB is subjected to thermal shock cycling, solder-float tests and vibration tests. Long-term damping for

humidity verification is done because the launch stations that the spacecraft are shifted to are typically located very close to the sea shore. The PCBs have to be well protected from the humidity point of view too. Hot and cold storage of PCBs is done, followed by vibration checks again. Lastly, a hot oil dip is done to find out if there is any lamination damage due to hot oil.

Some tests performed on spacecraft PCBs

Some of the tests that are performed on PCBs to ensure their high reliability are mentioned below, with a brief explanation on the purpose of each test.

Thermal shock cycling. The purpose of this test is to determine the resistance of a material, laminate or multi-layer board by repeated exposure to extreme temperatures for comparatively short periods of time. Interconnection resistance (ICR) is measured prior to and after thermal shock cycling. ICR must not change by more than 10 per cent of the initial value. Destructive tests like micro-sectioning are done to see whether the barrel or the plated through-hole is intact or not.

Thermal stress tests. This test is done to determine whether plated through-holes can withstand the thermodynamic effects of extreme heat to which they may be exposed during the assembly rework or repair process. The PCB is 'baked' at 125 to 150 degrees centigrade for six hours, after which it is subjected to six dips in the molten solder at 288 degrees centigrade, with a 120-second gap between the dips.

Vibration tests. This is an important event in the testing of PCBs because during vibration, anything can happen to the PCB. The interconnections are made through copper barrels that are coated inside the holes by chemical processes. Chemically deposited holes must be intact even after they are subjected to thermal cycling and vibration tests. The PCB is clamped on the 'vibration table' and sine input is swept ranging from 1 Hz